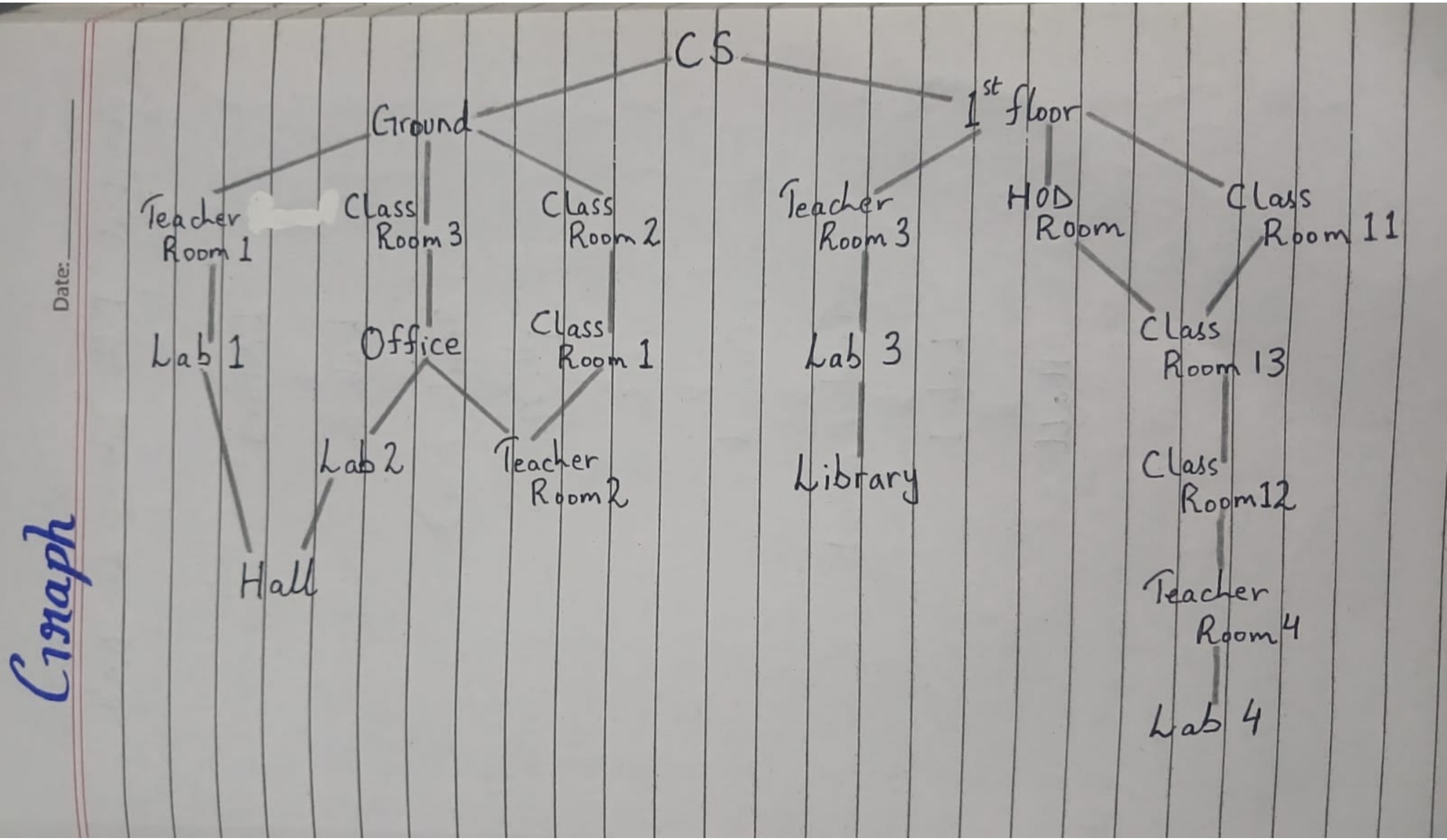
Date:

Graph


CS
Ground
1st floor
Teacher Room 1
Class Room 3
Class Room 2
Teacher Room 3
HOD Room
Class Room 11
Lab 1
Office
Class Room 1
Lab 3
Class Room 13
Lab 2
Teacher Room 2
Library
Class Room 12
Hall
Teacher Room 4
Lab 4

Date: _____

TRANSITION TABLE:

CS	→	Ground, 1st floor
Ground	→	Teacherroom 1, Classroom 3, Classroom 2, CS
Teacher Room 1	→	Ground, Lab 1
Classroom 3	→	Ground, Office
Classroom 2	→	Ground, classroom 1
Lab 1	→	Teacherroom 1, Hall
Office	→	Classroom 3, Lab 2, Teacher room 2
Classroom 1	→	Classroom 2, Teacherroom 2
Lab 2	→	Office, Hall
Teacher room 2	→	Office, Classroom 1
Hall	→	Lab 1, Lab 2
1st floor	→	Teacherroom 3, HODroom, Classroom 11
Teacherroom 3	→	1st floor, Lab 3
HOD room	→	1st floor, Classroom 13
Classroom 11	→	1st floor, classroom 13
Lab 3	→	Teacher room 3, Library
Classroom 13	→	HoDroom, Classroom 13, Classroom 11
Classroom 12	→	Classroom 13, Teacher room 4
Teacher room 4	→	Classroom 12, Lab 4
Lab 4	→	Teacher room 4
Library	→	Lab 3

Date:

Formulation of the Problem:

Initial State:
Library

Goal State:
Classroom 13

State space:
All accessible location in the CS department.

Action:
Move from one accessible state to an other adjacent state.

Transition model:
Purpose to move state

Path cost:
Distance.

Date:

BFS TABLE:

Initialization:
Q: [Library]
V: [Library]

deque Library & Expand
Q: [Lab 3]
V: [Library, Lab 3]

deque Lab 3 & Expand
Q: [TR 3]
V: [Library, Lab 3, TR 3]

deque TR 3 & Expand
Q: [1st floor]
V: [Library, Lab 3, TR 3, 1st floor]

deque 1st floor & Expand
Q [CS, HOD, CR 11]
V: [Library, Lab 3, TR 3, 1st floor, CS, HOD, CR 11]

Date:

dequeue CS & Expand

Q : [HOD , CR 11, Ground]

V : [Library, Lab 3, TR 3, 1st floor, CS,
HOD, CR 11, Ground]

dequeue HOD & Expand

Q : [CR 11 , Ground , CR 13]

V : [Library , Lab 3 , TR 3, 1st floor, CS,
HOD, CR 11, Ground, CR 13]

Date:

Shortest Path (Library - CR 13)

Library

↓

Lab 3

↓

Teacher Room 3

↓

1st floor

↓

HOD Room

↓

Class Room 13